Zecpath AI Technical Handbook

Overview

Zecpath AI is an end-to-end AI-powered hiring platform designed to automate the recruitment lifecycle from resume screening to final hiring recommendation.

System Modules

- Resume Parsing Engine
- ATS Scoring Engine
- Screening AI
- HR Interview AI
- Technical Interview AI
- Machine Test Evaluation
- Behavioral Analysis Engine
- Integrity Analysis Engine
- Cross-Round Aggregation Engine
- Decision AI
- Hiring Intelligence Report Generator

End-to-End Workflow

Resume Upload

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Resume Parser

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ATS Engine

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Screening AI

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HR Interview AI

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Technical Interview AI

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Machine Test Evaluation

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Behavioral Analysis

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Cross-Round Aggregation

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Decision AI

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Hiring Report

Core Features

- Resume Parsing
- Semantic ATS Matching
- Voice/Text Screening
- HR Interview Evaluation
- Technical Skill Assessment
- Machine Test Scoring
- Behavioral Analysis
- Integrity Detection
- Hiring Intelligence Report
- Recruiter Dashboard

Technology Stack

Backend

- Python
- FastAPI / Flask
- REST APIs

AI Libraries

- Sentence Transformers
- Scikit-learn
- Transformers
- NumPy
- Pandas

Database

- PostgreSQL
- MongoDB (Optional)

Deployment

- Docker
- Kubernetes
- AWS / GCP

AI Pipeline

Candidate

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Resume Parsing

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ATS Matching

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Interview Evaluation

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Behavior Analysis

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Score Aggregation

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Decision Engine

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Report Generation

Security

- JWT Authentication
- HTTPS
- Role-Based Access
- Audit Logs

- Input Validation

Monitoring

- API Logs
- AI Logs
- Error Logs
- Metrics
- Alerts
- Dashboard

Future Enhancements

- LLM-powered Interviews
- AI Video Analysis
- Emotion Detection
- AI Coaching
- Predictive Hiring
- Multi-language Interviews
- Continuous Learning Models

Version

Current Version: **v1.0**

Project: **Zecpath AI Hiring Platform**